WHEN TO WRITE A KUBERNETES OPERATOR

AND HOW TO DO IT RIGHT

(FEATURING: TEA)

ABOUT ME

Michael Schmitt

Sitereliability Engineer at interhyp.de

Inter*net* **Hyp***othek* (internet mortgage)

SO WHEN *DOES* AN OPERATOR MAKE SENSE?

When you need:

- relationships between resources
- invariants ("if X exists, Y must exist")
- lifecycle & cleanup
- continuous enforcement

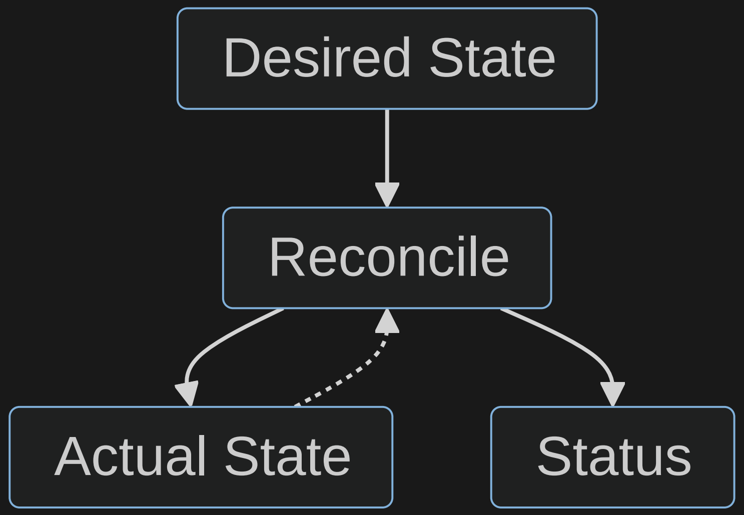
WHEN YOU *DON'T* NEED ONE

If it's:

- static resources
- one-off generation
- simple templating

➡️ **Just use YAML.**

NO MAGIC. JUST A LOOP.



WHAT MAKES A GOOD OPERATOR?

- simple reconciliation logic
- idempotent behavior
- clear, meaningful status
- boring, predictable outcomes

THE LINE TO DRAW

You declare

- What you want with YAML

The Operator handles

- All relationships
- Complete Lifecycle
- Keeps everything in desired state
- Tells you when it fails

LET'S SCAFFOLD AN OPERATOR...

1. Install Go, a Kubernetes Distribution, Get the Operator SDK

```
operator-sdk init --domain "$DOMAIN" --repo "$REPO"
```

```
operator-sdk create api \
  --group kitchen \
  --version v1alpha1 \
  --kind Teapot \
  --resource \
  --controller
```

DEMO

TAKEAWAYS

- Operators are powerful - and expensive
- Write them rarely
- Keep them small
- Status is your UX

THANK YOU :-)

- Presentation
- Golang
- Operator SDK
- Tea Pot Operator
- LinkedIn